



Particle Swarm Optimization on the Space of Quaternions with Applications to Attitude Estimation from Light Curves

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Abstract

This paper presents a novel multiplicative particle swarm optimization algorithm (MPSO) for optimization on the space of quaternions. It uses multiplicative quaternion kinematics to propagate quaternion particles, rather than the generic additive kinematics used in the traditional particle swarm optimization algorithm. The performance of the MPSO is demonstrated on the problem of attitude determination from light curves, which presents many challenges to traditional filtering-based attitude estimation approaches. The MPSO is shown to be able to successfully determine the attitude and angular velocity of an object from a set of multi-observer light-curve measurements. The MPSO is compared to other optimization approaches for this problem, and equal or superior performance is demonstrated on a variety of test cases. It is shown that the MPSO can be used to initialize conventional filters in order to successfully estimate attitude in situations where pure filtering would fail.

Keywords Optimization · Attitude determination · Quaternions · Light curves · Space situational awareness

1 Introduction

In the last few decades, space situational awareness (SSA) has become a critically important field for maintaining the safety of assets and humans in space. As the presence of humanity in orbit grows larger and larger, there is more and more risk of collisions that can cause catastrophic damage, even for collisions between relatively small objects and debris. According to the Executive Office of the President [1], in 2018 there were over 5,000 operational satellites in orbit, over 20,000

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objects larger than 10 cm and many times more small pieces of debris, in addition to the millions of objects too small to track. The task of observing and tracking all these objects is arduous and will only continue to grow more difficult. For this reason, new methods for tracking objects and debris must be developed that can scale with the ever-growing number of objects.

One of the major challenges that limits the ability to track debris is attitude dependent orbital perturbations such as solar radiation pressure (SRP) and drag. These effects cause orbits to decay or drift, and when un-modeled or poorly modeled result in discrepancies between prediction and reality, causing difficulty in associating new observations with old tracks, particularly when observations are sparse. The simplest models for SRP and drag are so called “cannon ball models” [2]. These models condense the geometry of an object into the drag and reflectivity coefficients, C_d and C_R , respectively, along with areas, A_{sun} and A_v , which are the cross-sectional areas of the object when viewed from the illumination direction and the velocity direction, respectively. These parameters are typically estimated from data, and often assumed constant for simplicity, despite being attitude (and thus time) dependent.

The simplifications introduced by these models create a substantial amount of error that degrades orbital predictions. In common applications, errors on the order of kilometers can build up in just a few days, and for high area-to-mass ratio objects, errors can build up to on the order of hundreds of kilometers in just a day or two [3]. Errors of these magnitudes can make measurement association arduous, and make it difficult to predict and avoid collisions. Models do exist that can more accurately predict SRP [4] and drag [5]; however, these models require knowledge of an object’s attitude and shape, which is frequently unknown in real systems where the user does not have direct control over the vehicle or object. For this reason, it is desirable to develop methods of obtaining the shape and attitude of objects externally so that more precise orbital predictions can be obtained from the higher-fidelity drag and SRP models, helping provide better collision avoidance predictions by providing better tracking of space objects where the observer has no intrinsic knowledge of the object’s attitude.

Conventional approaches to external attitude determination often utilize resolved images or radar [6]. However, these approaches are limited by range, and the required sensors typically have large size, weight and power footprints that limit their applications, particularly if they would need to be deployed on satellites. An alternative source of attitude information is light curves, which are non-resolved measurements, meaning they are only limited by their noise characteristics, and not by any angular resolution. They can be obtained from non-resolved imaging devices such as spectrometers, polarimeters [7], or from conventional optical images where the resolution is too coarse to resolve the object in question. These types of images are already generated from telescopes when obtaining the azimuth and elevation measurements needed for typical orbit determination, so this source of information is readily available. A method of externally determining attitude from light curves would allow this readily available source of information to be utilized to solve the pressing challenges of space-object tracking and conjunction analysis.

Approaches to identifying the attitude of an object can generally be divided into two categories [8], attitude determination and attitude estimation. Attitude determination is a “static” approach that takes measurements at a time point and determines an attitude at that time point. This requires that enough measurements are available at each point in time to uniquely identify the attitude. The second category, attitude estimation, generally refers to using sequential or batch filtering techniques to estimate an attitude. This paper primarily considers sequential filtering techniques because they are more relevant to real-time attitude estimation problems encountered frequently in space flight, but future work could include investigation of batch filtering as well. Attitude estimation via filtering has two advantages over static approaches: the first is that filtered attitude estimates can be provided efficiently in real time, and the second is that the attitude estimate can be updated from a single measurement source. The disadvantages of filtering techniques over static methods is that the filter may diverge if an inaccurate initial attitude estimate is given.

It will be demonstrated that, for this particular attitude estimation problem, the non-linearity of the light-curve model makes filtering approaches particularly sensitive to initial conditions, limiting their applicability to cases where the attitude is well-estimated in advance. For this reason, the following work focuses on developing approaches to attitude determination from light curves as a method of initializing attitude filtering approaches. This paper will show that effective attitude estimation via filtering can be achieved, given a sufficiently good initialization generated by attitude determination approaches. For static attitude determination, the minimum number of dimensions needed to parameterize attitude (three) necessitates that at least three different magnitude (or intensity) light-curve measurements from different observers must be provided to yield an attitude estimate. This is akin to requiring a minimum of, for example, a unit vector and an angle in standard attitude determination problems [9]. Classic gradient-based constrained optimization methods can be used to determine the attitude, but require a relatively good initial guess of the attitude in order to converge, meaning they do not perform better than filtering. In this paper, it is shown that particle swarm optimization algorithms are effective for solving the optimization problem without an initial guess, because they can better handle multi-modal optimization problems where many local minima exist.

In order to optimize on the space of attitudes, an attitude representation must be chosen. There are many attitude representations that can be used [10]. A particle swarm optimization approach to determine the attitude and other parameters is shown in [11]. However, the attitude is represented using three angles, which is a minimal representation, as a local representation of a global quaternion. It is well-known that all minimal representations have a singularity for certain rotations [12]. For this reason, the quaternion has become the standard representation for attitude determination. The quaternion is a four dimensional (non-minimal) representation, so it has no singularities, but instead has a unit-norm constraint that must be maintained. Furthermore, many typical optimization algorithms involve addition and subtraction operations on the design variables. In the case of quaternions as a design variable, addition is not closed on the set of unit quaternions, so this operation will not generally yield valid attitudes without additional constraints. To overcome these issues, this paper develops a multiplicative particle swarm optimization (MPSO)

algorithm, which is composed of analogous operations to the addition and subtraction in the standard PSO, so that optimization can be performed directly on the set of unit quaternions, and so that the unit quaternion constraint is implicitly enforced. The result is a novel optimization algorithm that can be used to determine attitude, and performs better than alternatives for highly nonlinear light curve models.

The organization of this paper is as follows. First, the definitions of the mathematical models utilized for attitude and light curves are given. This is followed by a discussion of the issues encountered when utilizing light curves as measurements in a filter. Then, the derivation of the MPSO algorithm is shown. This is followed by simulation results that show the effectiveness of the MPSO algorithm, including demonstration of filtering performance when initialized using the MPSO algorithm. Finally, conclusions are stated.

2 Modeling

Before the approach is presented, a number of models must be defined so that the optimization problem can be clearly stated. First, attitude representations will be discussed. Then a model of flat-facet object geometry is presented and a light-curve model is defined. Finally an optimization problem is constructed in order to solve the attitude determination problem from light curves.

2.1 Attitude Representations

The orientation of an object can be represented in many different ways. The following sections define several common representations and discuss their usefulness for the purpose of attitude determination from light curves.

2.1.1 The Direction Cosine Matrix

A common attitude representation is the direction cosine matrix (DCM), A . This is one of the most conceptually fundamental attitude representations, and can be constructed from the cosines of the angles between the axes of two orthogonal coordinate frames. It is a 3×3 orthogonal matrix, satisfying

$$A^T A = A A^T = I_{3 \times 3} \quad (1)$$

where $I_{3 \times 3}$ is the 3×3 identity matrix, and $()^T$ is the matrix transpose operator. The DCM contains nine total elements; however, these elements are not independent. Equation (1) is a constraint that reduces the number of degrees of freedom to three. DCMs can be used to easily transform vectors between reference frames. A DCM's column vectors are orthogonal by definition, as shown in Eq. (1). Multiplying a vector by the DCM transforms it into the reference frame defined by the columns of the DCM. While DCMs make transformations of vectors between frames very simple, they require nine parameters, meaning they are not ideal for computation compared

to other attitude representations, particularly in real-time systems where memory and computational cost are major design considerations.

2.1.2 Quaternions

A second option for representing attitude is the quaternion, represented by a four-element vector consisting of a three-element “vector part” and one-element “scalar part.” This can be notated as

$$\mathbf{q} = \begin{bmatrix} \mathbf{q}_{1:3} \\ q_4 \end{bmatrix} \quad (2a)$$

$$\text{with } \|\mathbf{q}\| = 1 \quad (2b)$$

where $\|\mathbf{x}\|$ is the Euclidean norm. Note that for the purposes of attitude representation, the set of unit quaternions has a 2:1 mapping, meaning that $\mathbf{q}$ and $-\mathbf{q}$ provide the same attitude. Also, the unit-length constraint in Eq. (2b) must be enforced to ensure that the quaternion represents a valid attitude. The relationship between the DCM and the quaternions is given by

$$A(\mathbf{q}) = (q_4^2 - \mathbf{q}_{1:3}^T \mathbf{q}_{1:3}) I_{3 \times 3} + 2\mathbf{q}_{1:3} \mathbf{q}_{1:3}^T - 2q_4 [\mathbf{q}_{1:3} \times] \quad (3)$$

where $[\mathbf{v} \times]$ is the cross-product matrix of the vector $\mathbf{v}$ [8]. A quaternion product between two quaternions ($\mathbf{q}$ and $\mathbf{q}'$) is defined as:

$$\mathbf{q}' \otimes \mathbf{q} = [\Xi(\mathbf{q}) \quad \mathbf{q}] \mathbf{q}' = [\Psi(\mathbf{q}') \quad \mathbf{q}'] \mathbf{q} \quad (4a)$$

$$\Xi(\mathbf{q}) = \begin{bmatrix} q_4 I_{3 \times 3} + [\mathbf{q}_{1:3} \times] \\ -\mathbf{q}_{1:3}^T \end{bmatrix} \quad (4b)$$

$$\Psi(\mathbf{q}) = \begin{bmatrix} q_4 I_{3 \times 3} - [\mathbf{q}_{1:3} \times] \\ -\mathbf{q}_{1:3}^T \end{bmatrix} \quad (4c)$$

A quaternion inverse is also defined as:

$$\mathbf{q}^{-1} = \begin{bmatrix} -\mathbf{q}_{1:3} \\ q_4 \end{bmatrix} \quad (5)$$

which satisfies the following properties:

$$\mathbf{q} \otimes \mathbf{q}^{-1} = \mathbf{q}_I = \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix} \quad (6a)$$

$$A(\mathbf{q}' \otimes \mathbf{q}) = A(\mathbf{q}') \otimes A(\mathbf{q}) \quad (6b)$$

where $\mathbf{q}_I$ is the identity quaternion and $A(\mathbf{q})$ is the attitude matrix corresponding to the quaternion $\mathbf{q}$ as defined in Eq. (3). The matrix $\Xi(\mathbf{q})$ also has several notable properties that will be useful for analysis:

$$\Xi^T(\mathbf{q}) \mathbf{q} = \mathbf{0} \quad (7a)$$

$$\Xi(\mathbf{q}_1 + \mathbf{q}_2) = \Xi(\mathbf{q}_1) + \Xi(\mathbf{q}_2) \quad (7b)$$

The quaternion is widely used for attitude representation in real systems because it is a lower dimensional representation of attitude than a DCM, while avoiding the singularity issues that are present with all three-dimensional attitude representations.

2.1.3 Generalized Rodrigues Parameters

A minimal parameterization is also needed for comparison and analysis. Generalized Rodrigues Parameters (GRPs) [13] are one such attitude representation, and are chosen here for several reasons that will be expanded upon later. GRPs are related to quaternions as follows:

$$\mathbf{p} = \frac{f \mathbf{q}_{1:3}}{a + q_4} \quad (8)$$

where a and f are arbitrary constants. The inverse relationship is given by:

$$q_4 = \frac{-ap_n^2 + f \sqrt{f^2 + (1 - a^2)p_n^2}}{f^2 + p_n^2} \quad (9a)$$

$$p_n = \|\mathbf{p}\| \quad (9b)$$

$$\mathbf{q}_{1:3} = \frac{(a + q_4)\mathbf{p}}{f} \quad (9c)$$

Note that in the case where $a = 1$, Eq. (9) simplifies to

$$q_4 = \frac{f^2 - p_n^2}{f^2 + p_n^2} \quad (10a)$$

$$p_n = \|\mathbf{p}\| \quad (10b)$$

$$\mathbf{q}_{1:3} = \frac{2f}{f^2 + p_n^2} \mathbf{p} \quad (10c)$$

There are several special cases of these constants. Setting $a = 0$ and $f = 1$ corresponds to the “classical Rodrigues parameters,” and setting $a = f = 1$, corresponds to the “modified Rodrigues parameters” (MRPs), given by

$$p = \frac{q_{1:3}}{1 + q_4} \quad (11)$$

The MRPs can also be seen as the stereographic projection of the quaternions onto a three-dimensional hyperplane. MRPs have a singularity for principal rotations of 0° or 360° . In addition, there exists a so-called ‘shadow set’ of MRPs defined as

$$p_s = \frac{-p}{p^T p} \quad (12)$$

Notably, the combination of MRPs and shadow MRPs form a complete covering of the attitude space within the the unit sphere; a fact that will be utilized later. This is primary motivation for choosing MRPs over other minimal representations, such as Euler angles.

2.2 Attitude Kinematics

Kinematics can be derived for any attitude representation; however, in the interest of brevity, only the quaternion kinematics are shown here because they will be used in the derivation of the MPSO. The general continuous-time quaternion kinematics are given as

$$\dot{q} = \frac{1}{2} \Xi(q) \omega \quad (13)$$

where ω is the angular velocity vector and $\Xi(q)$ is defined in Eq. (4b). It can also be shown [14] that this is equivalent to

$$\dot{q} = \frac{1}{2} \Omega(\omega) q \quad (14a)$$

$$\Omega(\omega) = \begin{bmatrix} -[\omega \times] & \omega \\ -\omega^T & 0 \end{bmatrix} \quad (14b)$$

Generally ω is the angular velocity vector that describes the rotation of the body frame with respect to the inertial frame, expressed in the body frame. For the purposes of developing kinematics for the particle swarm optimization algorithm, these two frames can be arbitrary. The inverse kinematics are given by

$$\omega = 2 \Xi^T(q) \dot{q} \quad (15)$$

Discrete forms of these equations are also derived for use in the MPSO. The forward kinematics in Eq. (13) have an exact discrete solution, given by [14]

$$q^{n+1} = \Omega_d(\omega) q^n \triangleq \text{Kin}(q^n, \omega) \quad (16a)$$

$$\Omega_d(\omega) = \begin{bmatrix} \cos\left(\frac{1}{2}\|\omega\|\Delta t\right)I_{3\times 3} - [\phi\times] & \phi \\ -\phi^T & \cos\left(\frac{1}{2}\|\omega\|\Delta t\right) \end{bmatrix} \quad (16b)$$

$$\phi = \sin\left(\frac{1}{2}\|\omega\|\Delta t\right)\frac{\omega}{\|\omega\|} \quad (16c)$$

where Δt is the discretization time step. Equation (16b) can also be written as

$$\Omega_d(\omega) = \cos\left(\frac{1}{2}\|\omega\|\Delta t\right)I_{4\times 4} + \frac{\sin\left(\frac{1}{2}\|\omega\|\Delta t\right)}{\|\omega\|}\Omega(\omega) \quad (17)$$

The inverse kinematics in Eq. (16) can be discretized by approximating the quaternion rates $\dot{q}$:

$$\dot{q} \approx \frac{q(t + \Delta t) - q(t)}{\Delta t} = \frac{q^{n+1} - q^n}{\Delta t} \quad (18)$$

This is a common approximation of the quaternion rate via a simple forward Euler formulation, that is valid for small changes in attitude. Substituting Eq. (18) into Eq.(15) gives

$$\omega = 2\Xi^T(q)\frac{q^{n+1} - q^n}{\Delta t} \stackrel{\Delta}{=} \text{Kin}^{-1}(q^{n+1}, q^n) \quad (19)$$

2.3 Object Geometry

In order to define a model of reflection, one must first define the object that is reflecting. Some models exist for specific curved surface geometries such as spheres and cylinders [15]; however, these models are not as easily generalized. For the purposes of SSA, many objects can be accurately represented by sets of flat facets. For instance, solar panels and rectangular satellite busses are easily represented in this way. For more complex geometries, the flat-facet assumption can easily be satisfied using paneling methods or meshing techniques. For this reason, the light curve model used in this research is developed under the assumption that the reflecting object is composed entirely of flat facets.

Figure 1 shows a diagram of a single facet along with a number of associated vectors needed to evaluate the reflection model. The flat facet is represented by a set of three orthogonal vectors: a normal vector u_n and two in-plane vectors u_u and u_v . The facet vectors are assumed to be known in the body frame of the object, denoted with a superscript B . In addition, unit vectors pointing from the object to the light source u_{sun} and to the observer u_{obs} are assumed known in the inertial frame, denoted with a superscript I .

Because these vectors are known in different frames, a frame transformation is required in order to evaluate the model in a common frame. Vectors in the inertial frame and body frame are related by the attitude matrix, A : a DCM that transforms

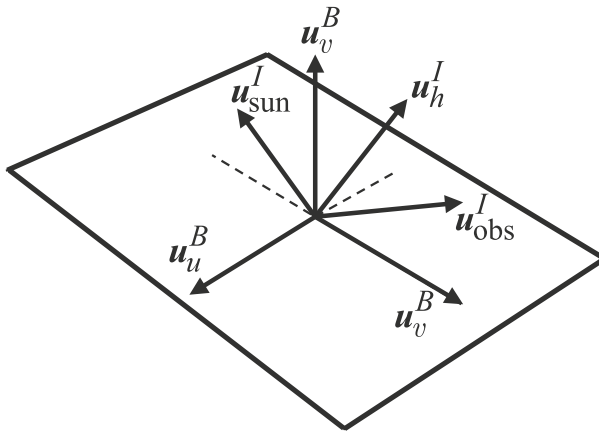


Fig. 1 Flat-facet geometry

vectors in the inertial frame into the body frame. By the definition of the DCM in Eq. (1), the inverse transformation from the body frame to the inertial frame is given by

$$\mathbf{v}^I = \mathbf{A}^T \mathbf{v}^B \quad (20)$$

The half-angle vector, denoted by $\mathbf{u}_h^I$, between the Sun and observer vectors is also used in many reflection models, and is given by

$$\mathbf{u}_h^I = \frac{\mathbf{u}_{\text{sun}}^I + \mathbf{u}_{\text{obs}}^I}{\|\mathbf{u}_{\text{sun}}^I + \mathbf{u}_{\text{obs}}^I\|} \quad (21)$$

2.4 Light Curve Model

With a model for flat facets defined, a reflection model can be developed. Reference [16] gives a general model of reflected intensity from a flat facet as

$$F_{\text{obs},ij} = \frac{C_{\text{sun}} \rho_{\text{total},ij} (\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{sun}}^I) \mathcal{A}_i (\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{obs},j}^I)}{d_j^2} \quad (22)$$

where C_{sun} is the solar intensity, $\mathbf{u}_{n,i}^I$, $\mathbf{u}_{\text{sun}}^I$ and $\mathbf{u}_{\text{obs},j}^I$ are unit vectors defined in Fig. 1, $\mathcal{A}_i$ is the facet area, d_j is the distance from the object to the observer, and $\rho_{\text{total},ij}$ is the bidirectional reflectance distribution function that will be defined later. The subscripts i and j denote the facet and observer, respectively. Material properties in the model along with C_{sun} are specific to a given frequency band on the electromagnetic spectrum. The parameter C_{sun} in particular can be approximated as a constant under the point source illumination assumption and given a frequency band on the

electromagnetic spectrum. Previous work [17–19] has used a value of $C_{\text{sun}} = 455 \frac{\text{W}}{\text{m}^2}$ for the visible light spectrum. The other parameters are known from the geometry of the object and the locations of the object and observers. There is also a line-of-sight constraint, requiring that

$$(\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{sun}}^I) > 0 \quad (23a)$$

$$(\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{obs},j}^I) > 0 \quad (23b)$$

The total intensity measured by the j^{th} observer is then the sum of the observed intensity from each facet:

$$F_{\text{total},j,\lambda} = \sum_{i \in \mathcal{F}} F_{\text{obs},i,j} \quad (24)$$

where $\mathcal{F}$ is the set of facets satisfying the line-of-sight constraints. Additionally, sometimes apparent magnitude is used instead of intensity, particularly in the astronomy and astrophysics communities where it allows for consideration of objects with orders of magnitude difference in intensity. The relationship between the intensity and apparent magnitude is given by

$$m_{\text{app}} = -26.7 - 2.5 \times \log_{10} \left| \frac{F_{\text{total}}}{C_{\text{sun}}} \right| \quad (25)$$

where -26.7 is the apparent magnitude of the Sun. This expression will not be used in this paper, but it is valuable to include it here to add context within the literature.

2.4.1 Bidirectional Reflectance Distribution Function

The bidirectional reflectance distribution function (BRDF) is a function that determines the fraction of light incident on a facet that is reflected in a specified direction. In general, there are no exact models, and a number of different models (both empirical and analytical) exist that have varying degrees of accuracy for different materials. For this work, the Ashikhmin-Shirley model [20] is selected. This model is a modification of the Blinn-Phong model [21], designed to eliminate the drawbacks that prevent the original Blinn-Phong model from being physically realistic. Namely, the Ashikhmin-Shirley model conserves energy [20]. The Ashikhmin-Shirley model includes a unique diffuse reflection term that is derived to ensure that the total reflected energy between the diffuse and specular terms never exceed the incident energy. The Ashikhmin-Shirley model also models non-isotropic reflections. The full model is given by

$$\rho_{\text{total},i,j} = \rho_{\text{spec},i,j} + \rho_{\text{diff},i,j} \quad (26a)$$

$$\rho_{\text{spec},i,j} = K \frac{(\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{h,j}^I)^\wedge \left[\frac{n_u (\mathbf{u}_{h,j}^I \cdot \mathbf{u}_{u,i}^I)^2 + n_v (\mathbf{u}_{h,j}^I \cdot \mathbf{u}_{v,i}^I)^2}{1 - (\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{h,j}^I)^2} \right]}{(\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{sun}}^I) + (\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{obs},j}^I) - (\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{sun}}^I)(\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{obs},j}^I)} F_{\text{ref}} \quad (26b)$$

$$\rho_{\text{diff},i,j} = \frac{28R_{\text{diff},i}}{23\pi} (1 - R_{\text{spec},i}) \left[1 - \left(1 - \frac{\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{sun}}^I}{2} \right)^5 \right] \left[1 - \left(1 - \frac{\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{obs},j}^I}{2} \right)^5 \right] \quad (26c)$$

$$F_{\text{ref}} = R_{\text{spec},i} + (1 - R_{\text{spec},i})(1 - \mathbf{u}_{h,j}^I \cdot \mathbf{u}_{\text{sun}}^I) \quad (26d)$$

$$K = \frac{\sqrt{(n_u + 1)(n_v + 1)}}{8\pi} \quad (26e)$$

The total BRDF is broken into two components: the specular and diffuse terms. Diffuse reflection is typically in all directions (although not necessarily uniformly), and specular reflection is concentrated around a particular direction. Diffuse reflection is typical of matte materials while specular reflection is often described as “mirror like” and strongly exhibited by metallic materials. The rest of the model is defined in terms of the previously defined unit vectors from Fig. 1 along with the material property parameters n_u , n_v , R_{spec} and R_{diff} . The parameters n_u and n_v define the spectral reflection cone around the direction of reflection by determining the size of the cone in the u and v dimensions. This allows for non-isotropic reflection. In the isotropic case, the exponent of the spectral reflection term reduces to

$$\frac{n_u (\mathbf{u}_{h,j}^I \cdot \mathbf{u}_{u,i}^I)^2 + n_v (\mathbf{u}_{h,j}^I \cdot \mathbf{u}_{v,i}^I)^2}{1 - (\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{h,j}^I)^2} = n_r = n_u = n_v \quad (27)$$

Additionally, the spectral reflection coefficient R_{spec} can be defined in terms of more common material properties:

$$R_{\text{spec}} = \frac{(n - 1)^2 + k^2}{(n + 1)^2 + k^2} \quad (28)$$

where n is the index of refraction and k is the extinction coefficient.

2.5 Self Shadowing

The above model assumes the entire area of each facet is visible; however, for non-convex objects, parts of the object could block the line-of-sight to other parts of the object, effectively reducing the visible area of the object. To account for this, the facet area can be made a function of attitude. To do this, the vertices of each facet are projected onto the plane normal to the light source or observer, and then a polygon clipping algorithm can be used to determine the shadowed area. Finally, the proportion

of shadowed area can be determined and the full area of the facet is reduced by that proportion. Pseudo-code for the algorithm is summarized in Algorithm 1.

Algorithm 1 Self-shadowing algorithm

```

1: for each visible facet do
2:   project vertices onto Sun-normal plane
3: end for
4: for each pair of visible facets do
5:   if projected facets overlap then
6:     perform clipping
7:   end if
8: end for
9: for each visible facet do
10:  project original and clipped vertices of facet onto observer-normal plane
11: end for
12: for each pair of visible facets do
13:  if projected facets overlap then
14:    perform clipping
15:  end if
16: end for
17: for each facet do
18:  if facet is visible then
19:    compute the modified area
20:  end if
21: end for

```

Projection can be done using Eq. (20), applied to each vertex of the facet to be projected. The exact rotation matrix used to perform the projection will depend on the specific projection, but in general, all of the necessary DCMs for projection can be determined from the Sun and observer directions, which are assumed to be known, along with the attitude of the object, which is an input to the model (and the design variable for optimization). Clipping of polygons can be done in a number of ways; however, for this work, the Weiler-Atherton clipping algorithm can be used for this purpose with minor modification to obtain the clipped polygon rather than the intersecting polygon. The reader can refer to [22] for more details on this algorithm. There are other phenomena such as multi-path illumination that could also be considered via techniques such as ray tracing; however, those considerations are outside the scope of this work.

2.6 A Simple Test Case

Given the models developed in the previous section, a simple test case is devised involving a number of observers looking at a satellite. The light curve model is used to generate measurements. For the purpose of filtering, intensity measurement uncertainty is assumed to be Gaussian with a standard deviation of approximately $1 \times 10^{-15} \frac{W}{m^2}$, and available at 1 Hz. One observer is used for filtering unless otherwise specified. The object is assumed to be freely rotating with an initial angular

velocity randomly sampled from a Gaussian distribution with 100 degrees per hour 1σ [23]. The moment of inertia of the object is approximated from the object geometry and material densities. For optimization, no measurement uncertainty is included in the interest of brevity. Results considering uncertain measurements are included in [23], which the reader can refer to for more details. Three observers are the minimum necessary for static attitude determination, but it was found [23] that a fourth observer substantially improves performance, so 4 observers are used unless otherwise specified. When angular velocity is determined via optimization, a second measurement is generated for each observer 0.1 s after the first measurement. The satellite is modeled via a basic box-wing geometry with dimensions loosely based on the Galaxy series of satellites [24]. This satellite has four major elements: a rectangular bus, a circular dish, and two solar panels. The chosen dimensions and material properties for each of these components are given in Table 1. The parameters n_u and n_v are both equal to 1,000. In addition, the unit vectors pointing to the Sun and observers are given in Table 2. The satellite model is shown in Fig. 2. This is equivalent to assuming the satellite is at a known location relative to the observer, and only the attitude is unknown. All future analysis will use these parameters unless otherwise specified.

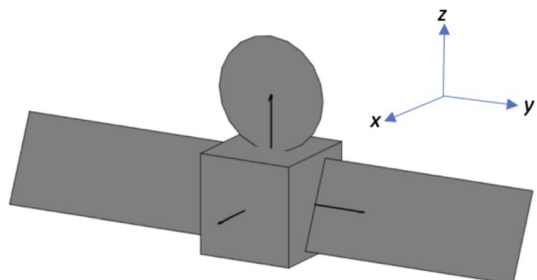
Table 1 Geometry and properties

Part	Dimensions (m)	Surface Properties
Bus	Height: 1.7, Width: 1.75, Depth: 1.8	$R_{\text{spec}}: 0.26, R_{\text{diff}}: 0.6$
Solar Panels	Length: 4 Width: 1.6	$R_{\text{spec}}: 0.04, R_{\text{diff}}: 0.05$
Dish	Diameter: 1.872	$R_{\text{spec}}^1: 0.275/0.26 \ R_{\text{diff}}: 0.275/0.6$

¹ Two values for the front and back of the dish

Table 2 Observer and sun vectors

	$\mathbf{u}_{\text{sun}}$	$\mathbf{u}_{\text{obs1}}$	$\mathbf{u}_{\text{obs2}}$	$\mathbf{u}_{\text{obs3}}$	$\mathbf{u}_{\text{obs4}}$	$\mathbf{u}_{\text{obs5}}$
Inertial Unit Vector	$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$	$\begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 0 \end{bmatrix}$	$\begin{bmatrix} \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{bmatrix}$	$\begin{bmatrix} \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{bmatrix}$	$\begin{bmatrix} \frac{\sqrt{2}}{2} \\ 0 \\ \frac{\sqrt{2}}{2} \end{bmatrix}$

Fig. 2 Box wing model

3 The Failures of Filtering

Before developing an optimization approach, it is valuable to consider the motivation for using optimization. Traditionally, attitude estimation is achieved via filtering, so a natural first attempt at utilizing light curves would be to simply consider them as measurements in a filter. For this work, the multiplicative unscented Kalman filter is used. The purpose of this paper is not to develop novel filtering approaches, so the reader should consult [25] for details of the filtering approach. For the following simulations, initial attitude error is varied to demonstrate the sensitivity to initial attitude, while initial angular velocity error is sampled from a distribution with 1σ of 5 degrees per hour. Measurements are obtained at a rate of 1 Hz from a single observer.

A number of filtering results are shown that demonstrate the issues arising from the traditional filtering approach. Firstly, it is important to establish that the problem of attitude determination from light curves is observable. Some analysis has been done investigating this topic [18], and while the results are not entirely conclusive, experimental results seem to indicate that the problem is observable, albeit highly sensitive to initialization errors. Under ideal conditions where an accurate initial attitude estimate is available, good filtering performance can be observed as seen in Fig. 3 where the error in the estimated attitude and angular velocity converge quickly and are well-bounded by the estimated 3σ uncertainties denoted by red dashed lines.

However, when sufficiently large initial errors are present, divergent behavior is observed as seen in Fig. 4. To understand the dependency on initial error, a simulation is performed, where 100 random true attitude states are generated, and for each true attitude state, 60 corrupted attitude states are generated with errors ranging from 0.1 degrees to 30 degrees. The filters are initialized with the corrupted attitude states and run for one hour of simulation time. The last 20 s of each filtering simulation are tested for convergence. For convergence, it is stated that the norm of the attitude estimate error vector should be less than one degree and the norm of the angular velocity error vector should be less than one degree per hour. Additionally, it is desired that the error between the estimate and truth should fall reasonably within the uncertainty ellipsoid corresponding to the estimated covariance. A simple way to test this is to check if each element of the state estimate is less than three standard deviations (as determined by the estimate covariance) from the true estimate. These conditions constitute a ‘success criteria’ for filtering which will be used for the rest of this work. Note that a different success criteria will be devised for optimization-based attitude determination later in the paper. The results of this filtering experiment can be seen in Fig. 5. It is apparent that with only one or two observers, initial attitude errors greater than about 5 degrees can degrade filtering performance to the point where the solution rarely meets the success criteria. Even with 3 observers, the success rate is low enough that it would be inadvisable to rely on this approach for an attitude solution. It is clear that some other approach must be devised.

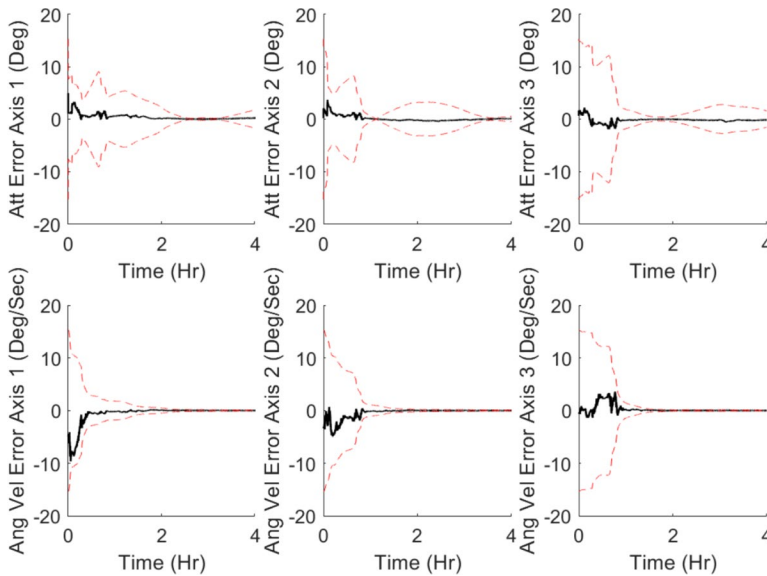


Fig. 3 Filtering error plot (5 degree initial error, 3σ uncertainties denoted by red dashed lines)

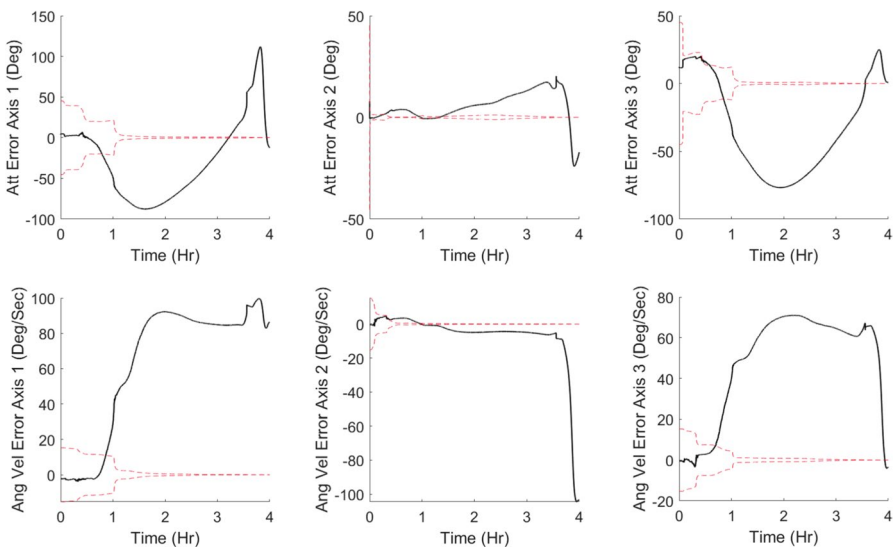


Fig. 4 Filtering error plot (15 degree initial error, 3σ uncertainties denoted by red dashed lines)

4 Optimization

Given a reflection model, an optimization problem can be posed such that solving the optimization problem is equivalent to determining the attitude of an object. Naturally, the design variables should be some representation of the attitude, so

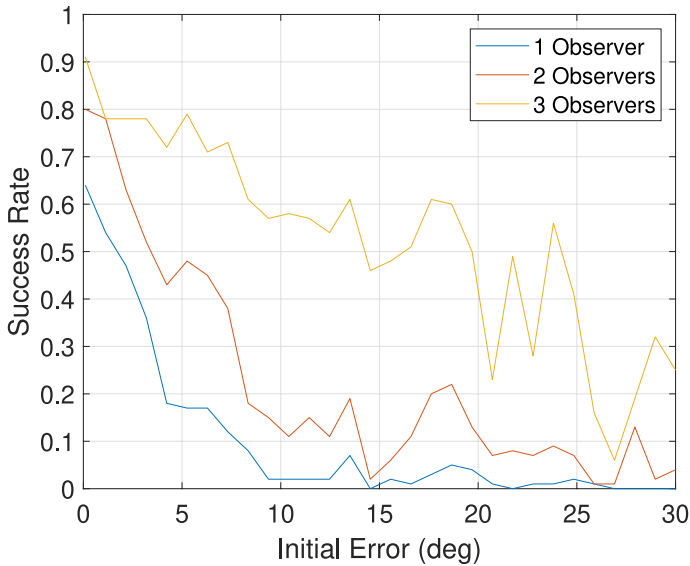


Fig. 5 Filtering success versus initial error

the cost function should give some metric that is zero only when the attitude is equal to the true attitude. In order for the attitude to be statically observable, at least 3 light curve measurements must be used [18], so the cost function should also be a function of a set of light curve measurements. A simple cost function is the sum of squared errors between the measurements, and a predicted measurement based on the model, given in Eqs. (24), (22), (26):

$$\min_{\mathbf{q}} \sum_j \frac{1}{\sigma_j^2} [\tilde{F}_{\text{total},j} - \hat{F}_{\text{total},j}(\mathbf{q})]^2 \quad (29)$$

where σ_j is the standard deviation of the measurement error associated with the j^{th} observer, and $\tilde{F}$ and $\hat{F}$ represent the measured and estimated intensities, respectively. For the purpose of simplicity, it is assumed that all observers have the same measurement noise characteristics, and thus $1/\sigma_j^2$ is a constant that scales the cost and can be safely ignored. In the future, it would be valuable to consider the measurement noise more rigorously. One major flaw with this simple cost function is that all of the measurement residuals are unweighted. Due to differences in distance between observers and the object, as well as the highly nonlinear nature of the cost function, the magnitudes of each measurement can vary by several orders of magnitude, and so unweighted residuals can lead to brighter measurements having a much larger impact on the cost function, which is not desirable. For this reason, a normalized cost function can be constructed in order to ensure each measurement is considered equally:

$$\min_q \sum_j \left[\frac{\tilde{F}_{\text{total},j} - \hat{F}_{\text{total},j}(\mathbf{q})}{\tilde{F}_{\text{total},j} + \delta} \right]^2 \quad (30)$$

where δ is a small constant to prevent numerical issues when the measurement is zero. A final consideration is the desire to estimate angular velocity in addition to attitude. This requires sequential measurements. The approach is to add angular velocity as an additional design variable and predict future measurements based on the angular velocity. Then the original cost function in Eq. (30) can be used to compute a cost at each measurement time and a total cost can be computed by summing these costs. The full cost function is given by

$$\min_s \sum_{n=0}^m \sum_j \left[\frac{\tilde{F}_{\text{total},j,n} - \hat{F}_{\text{total},j}(\mathbf{q}_n)}{\tilde{F}_{\text{total},j,n} + \delta} \right]^2 \quad (31a)$$

$$\mathbf{q}_{n+1} = \text{Kin}(\mathbf{q}_n, \boldsymbol{\omega}) \quad (31b)$$

$$s = \begin{bmatrix} \mathbf{q}_0 \\ \boldsymbol{\omega} \end{bmatrix} \quad (31c)$$

where the index n indicates the sequence of measurements over time, and the index j specifies the observer. The associated cost function is highly nonlinear as a consequence of the reflection model given in Eqs. (22) and (26), creating a large number of local minima in the cost function. For this reason a global optimization approach is necessitated. The following sections outline a novel global optimization technique: the multiplicative particle swarm optimization algorithm, which is derived specifically for optimization on the space of quaternions.

4.1 The Standard PSO Algorithm

The traditional particle swarm optimization algorithm is well-known in the optimization literature, and is notably a gradient-free global optimization algorithm [26–29]. The general concept is to initialize a number of candidate solutions called “particles” and then to allow those particles to move through the solution space based on some particle kinematics. Generally, these kinematics are designed to balance moving towards previously found good solutions while also continuing to explore new areas of the search space. The standard kinematics for a particular particle are given by

$$\mathbf{x}_i^{n+1} = \mathbf{x}_i^n + \mathbf{v}_i^{n+1} \quad (32a)$$

$$\mathbf{v}_i^{n+1} = \alpha \mathbf{v}_i^n + r_1 \beta_\ell (\mathbf{x}_i^{\ell, \text{opt}} - \mathbf{x}_i^n) + r_2 \beta_g (\mathbf{x}_i^{g, \text{opt}} - \mathbf{x}_i^n) \quad (32b)$$

Here r_1 and r_2 are random values in the range $r \in [0, 1]$, $\mathbf{x}_i^{\ell, \text{opt}}$ is the best solution found by this individual particle so far, $\mathbf{x}_i^{g, \text{opt}}$ is the best solution found in the global topology associated with the particle thus far, and β_ℓ , β_g and α are heuristic constants defined by the user that weight the different terms of the kinematics. The global particle topology is simply a system for determining which particles are influenced by each other [28]. A true global topology means $\mathbf{x}_i^{g, \text{opt}}$ is the best solution found by any particle.

Other topologies could involve each particle being influenced only by some small set of neighbors, or particles being assigned to clusters. The heuristic parameters can also be tuned to promote different aspects of exploration and convergence. Large α gives the particles “inertia” meaning they tend to continue moving in the same direction and explore more. Large β_ℓ causes the particles to stay near their individual best solutions, resulting in less exploration and more localized searching. Large β_g results in convergence towards the overall best solution found thus far. The PSO algorithm, like any gradient-free nonlinear optimization algorithm, can handle constraints via the penalty function method, but it is generally desirable to pose an unconstrained optimization problem. For this reason, the problem is formulated using modified Rodrigues parameters as the attitude representation, since a minimal representation can avoid constraints.

4.2 The Multiplicative PSO

The traditional PSO shown in Eq. (32) is a generic algorithm meant to be effective for any problem; however, a brief inspection of the particle kinematics reveals that they are not well suited to attitude optimization. Notably, they involve addition of the design variables. Adding attitudes together is generally not a useful operation, and may not even produce a valid attitude for some representations. In general attitudes are composed via products. For instance, the product associated with DCMs is matrix multiplication, and quaternions have their own product defined in Eq. (4a). The product of two attitudes generally results in a composite rotation through both attitudes.

For the purposes of particle swarm optimization, the kinematics involve computation of some velocity, and the propagation of the particles via that velocity. It then makes sense to develop equivalent equations for computing a “pseudo angular velocity” and propagating the attitude particles by that pseudo angular velocity. If the selected attitude representation is the quaternion, then Eq. (32a) can be replaced by the quaternion product of an update quaternion with the original particle. Equation (32b) is handled by computing equivalent terms as pseudo angular velocities that rotate the particle towards the local and global best solutions, and then adding those pseudo angular velocities together. Pseudo angular velocities can be computed using the inverse quaternion kinematics defined in Eq. (19). The resulting set of particle kinematics is given by

$$\omega_i^{\ell, \text{opt}} = \text{Kin}^{-1}(q_i^{\ell, \text{opt}}, q_i^n) \quad (33a)$$

$$\omega_i^{g, \text{opt}} = \text{Kin}^{-1}(q_i^{g, \text{opt}}, q_i^n) \quad (33b)$$

$$\omega_i^{n+1} = \alpha \omega_i^n + r_1 \beta_l \omega_i^{\ell, \text{opt}} + r_2 \beta_g \omega_i^{g, \text{opt}} \quad (33c)$$

$$q_i^{n+1} = \text{Kin}(q_i^n, \omega_i^{n+1}) \quad (33d)$$

This formulation has the added benefit of implicitly enforcing the quaternion normalization constraint, meaning the problem can be solved as an unconstrained optimization problem.

If the physical angular velocity is being determined in addition to the attitude, then hybrid particle kinematics must be used. This is because the multiplicative kinematics are only applicable to the attitude portion of the design variable vector. The hybrid system simply divides the design variable vector into attitude and non-attitude parts and then uses the MPSO kinematics for the attitude part, and the traditional kinematics for the non-attitude part. This is shown by

$$s_i^n = \begin{bmatrix} q_i^n \\ x_i^n \end{bmatrix} \quad (34a)$$

$$\omega_i^{\ell, \text{opt}} = \text{Kin}^{-1}(q_i^{\ell, \text{opt}}, q_i^n) \quad (34b)$$

$$\omega_i^{g, \text{opt}} = \text{Kin}^{-1}(q_i^{g, \text{opt}}, q_i^n) \quad (34c)$$

$$\omega_i^{n+1} = \alpha \omega_i^n + r_1 \beta_l \omega_i^{\ell, \text{opt}} + r_2 \beta_g \omega_i^{g, \text{opt}} \quad (34d)$$

$$v_i^{n+1} = \alpha v_i^n + r_1 \beta_\ell (x_i^{\ell, \text{opt}} - x_i^n) + r_2 \beta_g (x_i^{g, \text{opt}} - x_i^n) \quad (34e)$$

$$s_i^{n+1} = \begin{bmatrix} \text{Kin}(q_i^n, \omega_i^{n+1}) \\ x_i^n + v_i^{n+1} \end{bmatrix} \quad (34f)$$

4.3 Additional Considerations

As mentioned previously, a global particle topology needs to be chosen in order to implement the PSO or MPSO algorithms. For this work a cluster-based topology is used. This will be justified later in the results section. The k-means clustering algorithm [30] is a natural choice for this purpose. Analysis has also been done to investigate the dependence of optimization and computational performance on number of clusters [23]. In general, very small numbers of clusters (<10) yield poor performance, indicating that simple global topologies are not effective. Larger numbers of clusters than ~20–30 also do not yield any significant gains in performance, indicating that more localized topologies such as nearest neighbor topologies are not beneficial either. There was also minimal correlations between population size and required number of clusters, which is an interesting result. Other options for

clustering algorithms have also been considered, but it was found that there is minimal difference in performance between clustering algorithms, and k-means is simple to implement and fast to evaluate [23]. For this reason k-means clustering with 30 clusters is used to generate the results in this paper. It is possible that a better topology could be found, but the performance is insensitive enough to the variations tested that it did not seem worth pursuing further.

Additionally, some randomness can be introduced into cluster membership to further encourage exploration of the search space. This is achieved by introducing a parameter $\epsilon \in [0, 1]$. Every time clustering is performed, a random number in the range 0 to 1 is generated and if it is less than ϵ , then the particle is assigned to the cluster as determined by the clustering algorithm; otherwise, the particle is assigned to random cluster. This causes some small fraction of the particles to deviate away from their closest clusters and explore more of the search space.

Another minor modification to the basic PSO is the introduction of a cooling schedule for the coefficient of the inertial term α . This is implemented by making α a function of the iteration, so that it begins at a prescribed α_0 and decays linearly over time to a final prescribed value α_f . The purpose of this implementation is to not only ensure that the particles will converge in the final iterations towards the best solutions, but to also maintain a good level of exploration in the early iterations.

A final consideration is a “clean up” optimization routine. The concept is to first run a global optimization routine that is tuned to efficiently search for the general vicinity of the solution, but then to use a different optimization routine to converge down to the best solution as far as possible. In this case, a sequential least squares quadratic programming algorithm from the NLOpt optimization package [31] is found to be the most effective local optimization routine.

4.4 Other Optimization Routines

For the purpose of comparison, several other optimization routines are also considered from the NLOpt package [31]. Discussing the intricacies of these other optimization approaches is beyond the scope of this paper, but their names and citations are included in Table 3 and can be referenced if more details are needed. All these approaches are implemented in the NLOpt package [31], unless otherwise specified. Local algorithms are also divided in gradient-based (GB) and gradient-free (GF) algorithms.

Note that most of the global optimization algorithms require a local optimization algorithm to operate, and so pairs of local/global optimization algorithms are generally chosen. For this work, it is found that a small subset of algorithms substantially outperform the others, so results for most possible combinations are not shown. Additionally, when possible, a gradient-based local optimization routine is used. The gradient-free approaches are generally only considered when gradients of the cost function are not available, which occurs when considering self shadowing in this case.

A few unique optimization alternatives are also considered. One is an alternative formulation of the MPSO that involves replacing Eq. (33c) with the weighted

Table 3 Optimization Algorithms

Algorithm	Abbreviation	Type	Source
Controlled Random Search with Local Mutation	CRS	Global	[32, 33]
Stochastic Global Optimization	STOGO	Global	[34]
Multi Level Single Linkage Algorithm	MLSL	Global	[35]
Improved Stochastic Ranking Evolution Strategy	ISRES	Global	[36]
ESCH Algorithm	ESCH	Global	[37]
Low-Storage BFGS	LBFGS	Local (GB)	[38]
Sequential Least-Squares Quadratic Programming	SLSQP	Local (GB)	[39]
Shifted Limited-Memory Variable-Metric	SLMVM	Local (GB)	[40]
Method of Moving Asymptotes	MMA	Local (GB)	[41]
Local Optimization via Principal-axis Method	PRAXIS	Local (GF)	[42]
BOBYQA Algorithm	BOBYQA	Local (GF)	[43]
Subplex Algorithm	SBPLX	Local (GF)	[44]
NEWUOA Algorithm	NEWUOA	Local (GF)	[45]
Constrained Optimization by Linear Approximations	COBLA	Local (GF)	[46]

quaternion average of the local and global best solutions, along with the previous quaternion. The quaternion-averaging procedure is outlined in [47]. This will be referred to as MPSTO Alt. Additionally, a modified simulated-annealing approach (MSA) is considered that utilizes a quaternion specific notion of localization. The reader can see [23] for more details on this approach.

5 Results

In order to evaluate the performance of the various optimization routines, a series of tests are devised. All results will use the scenario outlined in Sect. 2.6, which assumes the satellite is at a known location relative to the observer, with known geometry and materials, and only the attitude is unknown. Investigation of the more general problem of simultaneous position and attitude estimation is beyond the scope of this work. Given this test case, metrics are defined in order to evaluate and compare performance. Notably, the goal of optimization in this case is to initialize filtering algorithms, so the performance metrics must be chosen to reflect that. Finally results are generated, both to compare optimization routines and to justify some design decisions that have been made.

5.1 PSO Heuristic Parameter Selection

The heuristic parameters for the PSO and MPSTO are chosen experimentally via parameter sweeps to identify the best performing values. In the interest of brevity, the results of the parameter sweeps are excluded, and the final chosen values are given in Tables 4 and 5.

Table 4 MPSO Selected Heuristics

β_ℓ	β_g	α_0	Topology	Alg	Pop. Size	Clusters	ϵ_0	Max. Iter
1.0	1.7	0.8	Cluster Based	k-means	600	30	0.7	80

Table 5 PSO Selected Heuristics

β_ℓ	β_g	α_0	Topology	Alg	Pop. Size	Clusters	ϵ_0	Max. Iter
1.4	1.5	0.5	Cluster Based	k-means	600	30	0.7	80

5.2 Optimization Performance Metrics

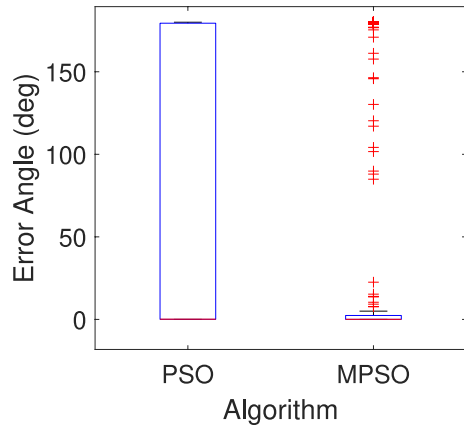
Typically, optimization-algorithm performance is measured by errors between the optimized solution and some true solution to the problem (if known). In this case a Euclidean error between vectors is not appropriate because the vectors in question are attitudes. Instead an attitude-error metric is defined given the true quaternion, $\mathbf{q}_{\text{true}}$, (known, but not used during optimization), and a quaternion generated by the optimizer (denoted $\mathbf{q} *$), the error-quaternion, given by

$$\mathbf{q}_e = \mathbf{q}_{\text{true}} \otimes \mathbf{q} *^{-1} \quad (35)$$

and the error metric is given by

$$\vartheta = 2 \arccos(q_{e,4}) = 2 \arccos(\mathbf{q}_{\text{true}} \cdot \mathbf{q} *) \quad (36)$$

A second problem with defining a performance metric is how to display results for large numbers of runs. Typically in the optimization literature, box plots would be used to represent the distribution of errors; however, that approach is somewhat ineffective for this problem due to the nature of the cost function. In particular, the cost function is very multi-modal, and the solutions tend to either converge very well, or fail to converge entirely, resulting in somewhat binary distributions of error that create very misleading box plots. This is highlighted in Fig. 6. The box plots show the 25th and 75th percentile of the data as the top and bottom of the box, and the red pluses are considered outliers. It can be seen that the data does not fit nicely into this format, and outliers are causing the box plots to look wildly different, despite the results being quite similar from the perspective of initializing filtering. For both algorithms, more than half of the solutions have less than 2 degrees of error, but that is nearly impossible to see from the plots. In the case of the PSO, almost no information about the actual performance of the algorithm can be gathered from the box plot. For this reason, an alternative must be pursued. Going back to the fundamental goal of initializing filtering, a success rate metric is proposed. This involves checking if the final error is within a success threshold and then categorizing each run as successful or failed. This makes sense in the context of filtering initialization, since the filter should be capable of converging with some moderate (a few degrees) error, so it is not particularly important for the optimization routine to converge as far as

Fig. 6 Box plot comparison

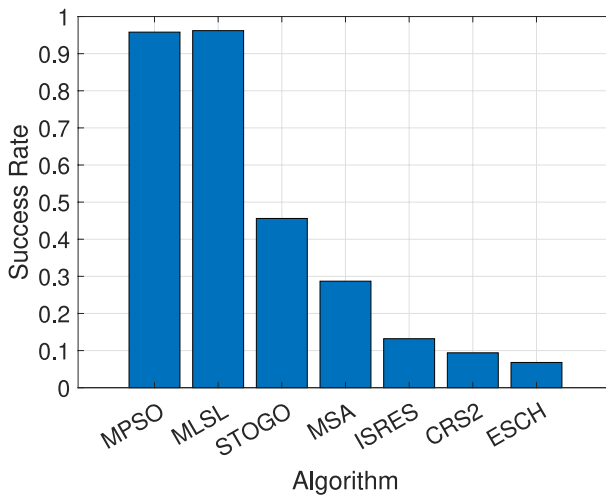
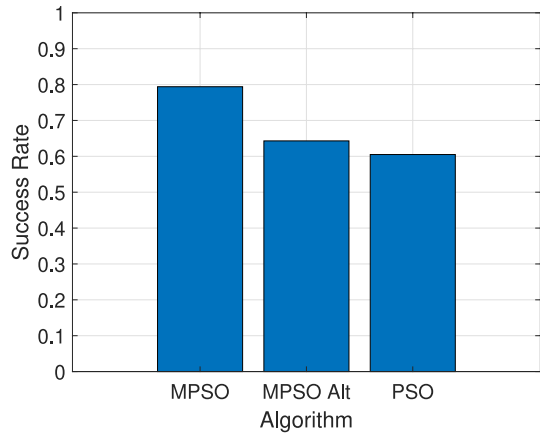
possible. From Fig. 5, a threshold of 5 degrees attitude error is chosen. Later results will also demonstrate that the success rate with this threshold corresponds well to filtering success rates.

5.3 Comparison of Optimization Algorithms

As an initial first pass, the standard light-curve model given in Eq. (22) is used with no self shadowing. This model is much faster to evaluate than the self-shadowing algorithm, making it practical to compare a large variety of algorithms. To begin, a simple comparison between the different variations of particle swarm optimization is performed. This includes the standard PSO, the MPSO, and the alternative formulation of the MPSO as defined in Sect. 4. For this test, no clean-up optimization is performed so that the comparison is as direct as possible. The results can be seen in Fig. 7. The MPSO substantially outperforms the alternatives, converging to the 5 degree threshold almost 20% more often than the standard PSO. For this reason, the remaining comparison will only consider the MPSO. Additionally the MPSO will be combined with a local clean-up optimization routine in the remaining comparisons.

In Fig. 8, performance is compared between the MPSO and number of alternative global optimization routines from the NLOpt package [31]. All optimizations are performed on the same hardware, so the performance of the algorithms can be compared directly. It can be seen that the MPSO achieves as good or better performance as every other algorithm, but there are alternatives that can achieve the same level of performance with this simpler cost function.

The advantage of the MPSO over other top performing algorithms like MLSL is most apparent when considering gradient-free optimization. In the case of self shadowing, gradient-free optimization is pursued because it is not possible to evaluate analytical gradients or perform auto-differentiation on the self-shadowing algorithm. Additionally, numerical differentiation is prohibitively computationally expensive on top of the already large computational burden of evaluating the self-shadowing algorithm for every cost function evaluation. In this case, the SLSQP gradient-based local optimization routine is replaced with a number of different gradient-free

Fig. 7 PSO dynamics success rate**Fig. 8** Comparison of global optimization approaches

optimization routines in combination with MLSL. The other algorithms from Fig. 8 are omitted from this experiment because their performance is poor enough that they would not reasonably compare to the top performing algorithms, and the substantial computational cost of evaluating many different algorithms would have been impractical within the available computational resources. The results of the MLSL with gradient-free local optimizers are compared to the MPSO as shown in Fig. 9. Here it can be seen that the MPSO substantially outperforms the alternative algorithms. In general, the addition of self shadowing makes the cost function much more expensive to evaluate, and the result is that algorithms take much longer (in real-time) to converge. It is believed that this gives the MPSO an advantage over the alternatives because it tends to more effectively search the global search space early on, giving it superior convergence properties in the early optimization iterations.

Investigating the specific reasons for the differences in performance between algorithms will be a topic of future investigation.

A final test is to consider the overall performance of a combined approach where filtering is initialized with the resulting attitude estimate from the optimization (in this case the MPSO). The point of this study is to show the efficacy of the filter when initialized using the estimates provided by the optimization routine. Figure 5 gives context to the performance of the filter with lower quality initializations. Here, results are shown using the higher quality optimized estimates for the filter. This validates the viability of optimization-based attitude determination to solve the filtering issues (outlined in Sect. 3) that motivated this work. These results are also necessary to validate the 5 degree threshold that is chosen to indicate successful convergence for the optimization routine in Sect. 5.2. A plot containing the success rate of optimization and the subsequent filtering is shown in Fig. 10. Success rate for the optimization is defined as convergence within the 5 degree threshold, and success rate for the filter is defined (as in Sect. 3) as convergence to within 1 degree for attitude and 1 degree per hour for angular rate, along with state estimate errors remaining with the estimated 3σ covariance ellipse. The optimization results are generated using measurements from 4 observers, which are used as initial estimates for the filters, but the filtering itself is done with 1, 2 or 3 observers in order to evaluate the sensitivity of the approach to varying numbers of observers. From these results, it can be seen that with only 2 observers, the filtering achieves essentially the same success rate as the optimization routine, and with one observer, there is only a minor drop off in success rate. This indicates that the filtering problem with 2 observers is primarily limited by the initialization accuracy. Additionally, the 5 degree threshold seems to effectively encompass successful initialization attempts since the success rate of the filter in converging stably seems to be the same as the rate of the optimization in achieving the specified 5 degree initialization accuracy.

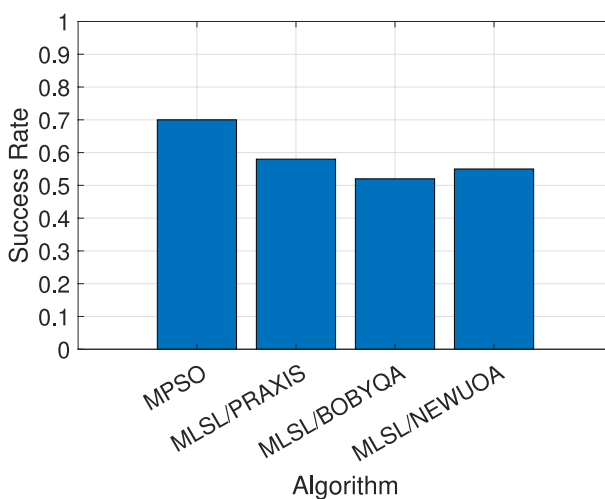


Fig. 9 Optimization performance (self shadowing)

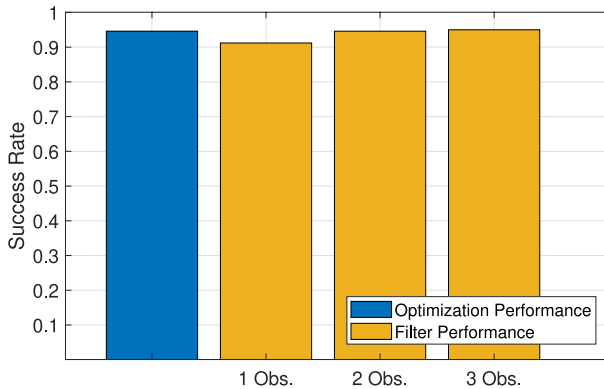


Fig. 10 Combined optimization and filtering performance

6 Conclusions

In this paper, the problem of attitude estimation from light curves is presented, and filtering results are shown to demonstrate the challenges of solving this problem conventionally. An alternative solution is devised to initialize filtering using attitude determination approaches, particularly an optimization routine that utilizes measurements from multiple observers to generate a static attitude and angular velocity estimate that can be used to initialize sequential filtering. For this purpose, a novel particle swarm optimization algorithm is derived for optimization on the space of quaternions. This algorithm utilizes quaternion kinematics to propagate quaternion particles effectively while simultaneously and implicitly satisfying the quaternion norm constraint. It is shown that this optimization routine is able to successfully initialize the filtering simulation with $>95\%$ success rate and that this optimization routine also matches or outperforms all considered alternative optimization approaches. The MPSO also demonstrates superior performance in the case where a self-shadowing model is considered, due to its ability to quickly search the design variable space, and to effectively handle arbitrary nonlinear cost functions. In the future, more work should be dedicated to higher fidelity modeling that can account for more phenomena such as multi-path scattering that are not considered in this work. This will lead towards an approach that could be effective in a real-world system.

Data availability Data is not available for this paper.

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